

Mobile Money and Mobile Technologies: A Structural Estimation

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Abstract. Using a data set on mobile technologies and mobile money in the emerging markets from 2000 to 2014, we examine the demand patterns of mobile technologies and mobile money when multiple generations of mobile technologies coexist in the market and each generation of the technologies may be bundled with mobile money. Using a structural model, we estimate the own and cross price elasticities for both mobile technologies and mobile money, and the demand effects between mobile money and mobile technologies. We find that the current, dominant technology (i.e., 3G) tends to have more robust demand as compared with other technologies that are either declining (i.e., 1G and 2G) or new to the market (i.e., 4G), and that customers are more likely to substitute forward toward newer technologies than backward toward older technologies when the price increases for a technology. We also find that mobile money differentiates the market and mitigates competition for those firms that offer mobile money. Theoretical and managerial implications are also discussed.

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Keywords: mobile money • mobile technology • demand estimation

1. Introduction

Mobile money is a technological innovation that combines mobile technologies and financial services to provide convenient and affordable access to banking systems and to improve financial inclusion (Lal and Sachdev 2015). It “allows users to deposit money onto their telephone handsets, transfer e-money to another user with a simple text message, and withdraw cash at one of thousands of outlets throughout the country” (Mbiti and Weil 2013, p. 369). One of the first mobile money services, M-PESA, was introduced in Kenya by the mobile network Safaricom and has been “by far the most successful example of mobile money banking in Africa” (Economist 2009). Its success has motivated continued adoption of mobile money in other developing countries (Jack et al. 2013, Jack and Suri 2014, Lal and Sachdev 2015, Mbiti and Weil 2013). By the end of 2017, a total of 276 mobile money services have been offered in more than 90 countries (GSMA 2017), and there are about 2.5 million mobile service agents serving consumers worldwide, with the network growing at 25.2% annually (Suri 2017). Mobile money is mostly adopted in developing countries with 49.1% of 690 million customers in Sub-Saharan African and 34% in South Asia (GSMA 2017).

The earliest success of mobile money was largely based on rapid developments of mobile technologies that enabled customers to access basic services such as airtime top-up, cash in and cash out, and domestic person-to-person (P2P) payments.¹ As mobile money evolved, mobile network operators (MNOs) attempted to bundle more value-added services to differentiate themselves in the market (Hughes and Lonie 2007). Consumers are now able to use mobile money to pay their bills (such as utilities), to make person-to-business (P2B) payments, to receive payments (such as wages) from business, to receive government-to-person (G2P) payments, or to use insurance and credit services (Donovan 2012, Suri 2017). GCASH in Philippines, for example, developed bulk payment services that enabled business customers to make salary disbursements through mobile money accounts, which cuts GCASH administrative and operational expenses from dealing with large numbers of in-person cash payments (Lal and Sachdev 2015).

An important aspect of mobile money is that the service is enabled by and has evolved with different generations of mobile technologies, to connect to external platforms, such as banking systems and payment provider systems, and to achieve full commercial functionality.²

Mobile money services hence require different levels of technical support, which can be facilitated or constrained by the technology platform. Specifically, first-generation mobile technology (1G), which was introduced in early 1980s provided the most basic mobile communications by introducing voice services based on analog transmissions. The second-generation (2G) emerged in late 1980s and increased the voice capacity and extended mobile access to the masses by leveraging different techniques such as Global System for Mobile Communications (GSM). Third-generation (3G) technology, launched in 2000, introduced broadband mobile access with data optimization, allowing high-speed internet access and also access to high quality audio, video, and new services such as Global Roaming (Qualcomm 2014). Fourth-generation (4G) technology, which was introduced in 2010, delivers more capacity for faster and better broadband access and for additional services such as dynamic information access and wearable devices (Qualcomm 2014). From voice and text messages to web browsing, and to fully connected intelligent sensors and devices, mobile technologies play an important role in mobile money service by blurring geographical and cultural borders.³ The evolution of mobile technologies highlights the consistent improvement in capacity and new functionality. For instance, the adoption of 3G technology significantly increased functionality, motivating rapid growth of mobile apps, whereas the adoption of 4G boosted data capacity for better service performance (Shrier et al. 2016).

Hence, mobile technology and mobile money represents a technology-service bundle that provides convenience and business value to customers (Jack and Suri 2014). The continued increase in mobile capacity and functionality during the evolution of mobile technologies may intensify or weaken market competition, and may complicate the impact of mobile money and service bundling on market performance of MNOs (Porter and Heppelmann 2014). This is because mobile money services require different levels of technical support, which can be facilitated, or constrained, by the technology platform on which any service is offered. As technologies advance quickly, the dynamics in continuous innovations may challenge the effectiveness of bundling strategies in offering mobile money services. For instance, certain technology platforms may be inadequate to support effective operations of certain mobile money services, and MNOs may overestimate their internal capabilities to offer such functionality.⁴ Hence, it has been a challenge for MNOs to optimize their pricing schemes and align their service offerings under evolving technologies, as the value proposition may change for customers when new capabilities and

options are available. Our research is motivated by this challenge and focuses on how the changes in mobile technology may affect the proliferation of mobile money. In particular, we address the following research questions: How do consumers choose mobile technologies when multiple generations of technologies coexist in the market? How do the provisions of mobile money and different generations of mobile technologies affect each other in demand?

A growing body of related literature has been on mobile banking (e.g., Xu et al. 2018), mobile payment (e.g., Mallat 2007, Xu et al. 2018), or mobile apps (e.g., Ghose and Han 2014) that focuses on their impact on sales, prices, and consumer surplus (Greenstein 1996, Brynjolfsson et al. 2003, Goolsbee and Petrin 2004, Narayanan et al. 2005, Mbiti and Weil 2013). Recently, a number of researchers have studied mobile money specifically (Jack and Suri 2014, Economides and Jezierski 2017). Although similar, mobile money is different from mobile banking and mobile payment in important ways. Mobile banking refers to the extension of online banking to mobile devices (Lal and Sachdev 2015), and mobile payment is an add-on service on mobile technology to facilitate fund transfers between individuals and/or merchants (Aron 2017). For both mobile banking and mobile payment, consumers are required to have a bank account in the back end. For mobile money, customers typically do not need bank accounts in the back end. Hence, the costs to access the service and switch between service providers are significantly lower than mobile banking and mobile payment, which has theoretical implications for consumer demands. Furthermore, the mobile technologies and mobile money are bundled together and affect each other in demand. The effect may come from within the different levels of technologies as the four-generation mobile technologies coexist in the market and act as substitutes. The strategy and outcome of bundling depend on marginal costs, valuation symmetry of individual products or services, and network effects such as network externality (Dewan and Freimer 2003, Prasad et al. 2010, Pang and Etzion 2012, Bhargava 2013, Etzion and Pang 2014).

We intend to narrow the literature gaps by examining the demand patterns of mobile technologies and mobile money. Using a data set on mobile money and mobile network operations in the emerging markets compiled from multiple data sources, we estimate the own- and cross-price elasticities for both mobile technologies and mobile money, and the demand effects between the mobile money and technologies. We also expand the estimations to other bundled financial services in addition to mobile money such as G2P, cross-sell, micro-insurance, and international P2P (interP2P hereafter).

Our key findings are that: (1) Although the own-price elasticities of all technologies are elastic, the

own-price elasticity for 3G technology is lower than the others, suggesting that current, dominant technology tends to have more robust demand as compared with other technologies that are either declining (i.e., 1G and 2G) or new to the market (i.e., 4G). The cross-price elasticities are higher for the effect of a technology on a newer technology than on an older technology, suggesting that customers are more likely to substitute forward toward a newer technology than backward toward an older technology when the price increases for the technology. (2) The own-price elasticity with mobile money is lower, in magnitude, than that without mobile money, suggesting that mobile money differentiates the market and mitigates the competition for those MNOs that offer mobile money service. In addition, the change in price with mobile money has greater impact on the market share of the service without mobile money than the change in price without mobile money has on the market share of the service with mobile money. (3) The magnitude of the own-price elasticity without mobile money for 3G technology is the lowest as compared with those for other generations of technology, likely because 3G is the current, dominant technology. Once mobile money is included, the elasticities are at a similar level for all generations of technology.

Our paper makes several important contributions. First, financial technology has gained much attention among practitioners and researchers alike. As a form of financial technology, mobile money has been widely and increasingly used in the emerging markets but under-studied by researchers, although a body of literature has been growing on mobile banking and mobile payments. It is important to understand consumer demand patterns associated with mobile money. Among the limited number of studies on mobile money, most of them are on the user-level and country-level economic issues associated with successes and failures in mobile money adoption (e.g., Jack et al. 2013, Mbiti and Weil 2013, Lal and Sachdev 2015, Economides and Jeziorski 2017). The paucity of literature on the demand and market competition of mobile money points our contributions. Second, mobile money is bundled with mobile technologies. The coexistence of multiple generations of mobile technology provides an interesting research setting to study the substitution patterns between different generations of mobile technology and the interactions between mobile money and mobile technologies. The existing literature on estimating the demand for mobile technologies is scarce, especially when multiple generations of mobile technology coexist as substitutes. A deeper understanding of the demand patterns for mobile technologies helps advance our understanding of emerging, innovative financial services such as mobile money. Our study contributes to the literature by examining the effects of

mobile money on consumer demand, conditional on the fast-evolving mobile technologies (from 1G to 4G), and by offering service bundling implications between mobile money and financial services.

The remainder of this paper is organized as follows. Section 2 presents a review of the background and related literature. Section 3 describes our research setting, data, and methodology. Section 4 presents our analyses, results, and robustness checks. Finally, Section 5 concludes the paper with discussion on findings, contributions, and future studies.

2. Background and Literature

In this section, we discuss the background of mobile money and the multiple streams of literature relevant to our research. Specifically, our paper relates to the literature on mobile money, bundling of services and technologies, and demand estimation using aggregate data.

2.1. Mobile Money

In recent years, mobile money has become a commonly adopted financial tool in many developing economies, allowing mobile phone users to deposit, transfer, and withdraw funds without formal bank accounts (Jack and Suri 2011). As mentioned earlier, mobile money is different from mobile banking. Mobile banking is defined as using a mobile phone to access bank account, either by accessing the bank's web page through the web browser on a mobile phone, via text messaging, or by using an app downloaded to the mobile phone (Federal Reserve Board 2013). Mobile money is a form of electronic money allowing the conduct of transactions through a mobile phone. Hence, it is typically provided by telecommunications companies, which is outside the formal banking system. For customers, mobile money is a menu item on their SIM card and allows them to engage in a variety of financial transactions such as payment on their mobile phone. Since bank accounts are not required, consumers need to go to a mobile money agent to register an account and deposit cash into the account (i.e., the mobile money). Once the mobile money is available, they can use their mobile phone to transfer that money to or receive money from anyone else in the country with a cell phone via their phone number. To cash out, they have to return to the agent who will cash the mobile money. The growth of agents is the key to the success of mobile money. Although not considered financial institutions, the agents act like financial institutions to help customers to cash in and cash out so that they can use mobile money.

The related literature in mobile money has mostly been organized into three streams: successes and failures in mobile money adoption, adoption patterns across different levels, and the impact of mobile money

on individuals, local communities, and countries. In the first stream of literature, scholars suggest that the success of mobile money systems depends on many factors, with agent network being the most important one (Suri 2017). Investment in agent networks with training and monitoring is crucial to achieve scale in offering mobile money services (Aron 2017). The demand for mobile money services is also affected by agent quality and competition (Balasubramanian and Drake 2015). Aside from the agent network, scholars also find that strong branding, service options, retail infrastructure and pricing, network penetration and coverage, and regulatory environment all play an important role in successful adoption of mobile money services (Heyer and Mas 2009, Mas and Morawczynski 2009, Lal and Sachdev 2015, Economides and Jeziorski 2017). For instance, Lal and Sachdev (2015) compare five successful deployments of mobile money to five less successful ones and highlight the importance of operating with a reliable mobile network and with a trusted brand and business. The second stream of literature studies the adoption patterns of mobile money. Specifically, scholars have investigated characteristics at the household level (e.g., household demographics), firm level (e.g., business size and transaction volume), and country level (e.g., regulatory environment) (Jack and Suri 2011, Oritgão 2015, Aron 2017). The last major stream of literature examines the impact of mobile money across various levels. Mobile money services can provide great reduction in transaction costs and time, as well as improvements in convenience and security (Economides and Jeziorski 2017). Jack and Suri (2014) find that a successful launch of mobile money, such as M-PESA, can empower households to share risks, whereas Suri and Jack (2016) suggest that better access to mobile money services has resulted in increases in household consumption and savings, and thus decreases in poverty rates. Dong et al. (2018) find that offering bundled mobile money services can lead to different impacts on the various value chain members (MNOs, banks, and end users).

Although there is a growing interest in academic research to investigate mobile money, most has focused on user-level or country-level issues that affect mobile money adoption. However, as mobile money has been bundled with mobile technology (i.e., mobile money is enabled by different generations of mobile technologies), a noticeable gap in the literature exists in understanding the challenges in bundling the mobile money and mobile technology in terms of demand patterns and pricing schemes. Our study intends to narrow the literature gap.

2.2. Bundling of Services and Technologies

Mobile money is a big business, but it is not easily profitable. By 2011, only M-PESA had achieved

operational profitability (Aron 2017). The success of mobile money depends on the value of the services offered to end-users, and the enhanced value comes from the effectiveness of service bundles. Product and service bundling creates revenue and reduces churn (Prince and Greenstein 2014), and it can also help differentiate a firm in a competitive environment (Stremersch and Tellis 2002). Service and technology bundles allow a firm to take advantage of complementarities from service and technological innovations (Milgrom and Roberts 1990). Yet the effectiveness of product and service bundles depends on many factors, one of which is value chain member coordination. For instance, when bundling involves various channel members, conflicts in the channels can weaken the incentives for bundling (Bhargava 2013). Dong et al. (2018) find that offering bundled mobile money services increases profits of banks, while it decreases profits of MNOs.

Technology development is another important factor, but has received less attention in the literature. Technologies, particularly wireless technologies, are particularly important in mobile money services as they can provide connectivity, protect secure data, provide reliability in service, and encourage product and service innovation for users across the globe. The popularity of mobile services is largely dependent on wireless technology systems. For example, technology companies are strengthening mobile money's ability to operate with online payment systems. Webtribe's Jambopay, Symbiotics' Moca, Intrepid Data System's iPay, and Pesapal are a few examples of payment services that allow buyers to pay for goods and services over the internet using their phones (Kendall et al. 2011). Another example is near field communications (NFC), which is a technology that allows devices to communicate through mere proximity by waving a specially equipped phone or card near a receiving device, as opposed to having to physically swipe it (Donovan 2012). NFC, enabled by 4G and potentially boosted by 5G, could serve to make transactions more efficient and secure by reducing errors, such as those that arise from mistyped numbers. In the coming years, more phones will be equipped with NFC, and for mobile money, this means that transactions can be completed by waving a phone near a receiver, as opposed to having to text value to a recipient. However, it has also been found that numbers of high-profile mobile money services were unable to scale up and meet the capacity demands of increasing numbers of consumers.⁵ Without understanding the dynamic changes of mobile technology systems on user demand for mobile money services, one cannot fully understand the opportunities and challenges MNOs face when bundling mobile money services under different technology systems. Hence, to fill the literature

gap, we examine the role of service bundling, and in particular, how mobile technology affects consumer demand for bundling.

2.3. Demand Estimation of Bundling: A Structural Estimation Approach

Demand estimation in differentiated markets has been studied in depth in economics, marketing, and information systems literature (e.g., Berry 1994, Berry et al. 1995, Kadiyali et al. 2001, Narayanan et al. 2005, Ghose and Han 2014), and there is a wide body of literature that documents the models of demand estimation. The aggregate demand estimation model can be estimated using market-level price and quantity data, and it handles the endogeneity of prices (Nevo 2000). The wider literature on demand estimation using this method also estimate demand for new product categories, such as mobile applications, under different industries such as automobiles, computers, books, and cable TV (Greenstein 1996, Brynjolfsson et al. 2003, Goolsbee and Petrin 2004, Narayanan et al. 2005, Ghose and Han 2014). More recently, Guajardo et al. (2016) use this method to study the effects of various product and service attributes on the demand for automobiles in the U.S., and McKie et al. (2018) employ the aggregate demand estimation model to study the impact of product and seller characteristics on the demand for refurbished products in the presence of new and used products. Similar to these studies, we leverage the Berry (1994) method for demand estimation and add to the literature by estimating the impact of price and wireless technologies and bundling characteristics on demand for mobile money services, using aggregate data.

3. Data and Methodology

3.1. Research Setting and Data

The research setting is the emerging markets for mobile money services. MNOs are the service providers of mobile money. The markets are bounded by countries; that is, a country is a market where multiple MNOs compete in. For example, in Kenya, four MNOs compete in the country by offering mobile money services: Airtel (Bharti Airtel), Orange (Telkom Kenya), Safaricom, and yuMobile (Essar Telecom). MNOs also offer different generations of mobile technology including 1G, 2G, 3G, and 4G. The mobile money services can be used with any generation of the technology. Consumers choose the generation of mobile technology and whether to use mobile money or not. Mobile money services include basic service items such as bill pay, domestic P2P, and additional mobile financial services such as G2P, InterP2P, cross-sell, and microinsurance.

We compiled a data set from multiple data sources including proprietary data provided by a market research

company and public data published by the World Bank. The market research company specializes in the telecommunication industry and operates mobile money deployment trackers to monitor global mobile money services. The first data set is the information collected from the mobile money deployment trackers that contains the MNO's names, the mobile money service's brand names (e.g., M-PESA in Kenya), the mobile money's launch dates, technology platforms (1G, 2G, 3G, or 4G), and additional financial services including G2P, InterP2P, cross-sell, and microinsurance. The second data set contains quarterly prices and market information of MNOs (e.g., average revenue per user (ARPU) and total connections) between 2000 and 2014. The databases include 407 MNOs from 79 countries.⁶ We also collected gross domestic product (GDP) and population data for all countries from the World Bank database to calculate GDP per capita.

To better understand our data, we constructed several figures. Figure 1 shows the average total connections and average ARPU per MNO in the data. Clearly, from 2000 to 2014, the mobile technology has been increasingly used and the price as measured by ARPU has continued to drop, likely due to heightened competition. Figure 2 shows the number of MNOs that offered mobile money and bundled services. Although the bundled services (G2P, InterP2P, cross-sell, and microinsurance) have remained relatively stable over time, mobile money has taken off with a rapid growth rate since 2008. Figure 3 presents the total connections percentage by the generation of mobile technologies. It can be seen that 2G is the most commonly used mobile technology, 3G and 4G are increasingly adopted, and 1G is declining. Figure 4 shows the percentage of users with access to mobile money (and not) in the top 10 countries. Brazil has the lowest percentage of mobile money users,

Figure 1. (Color online) Average Total Connections and ARPU

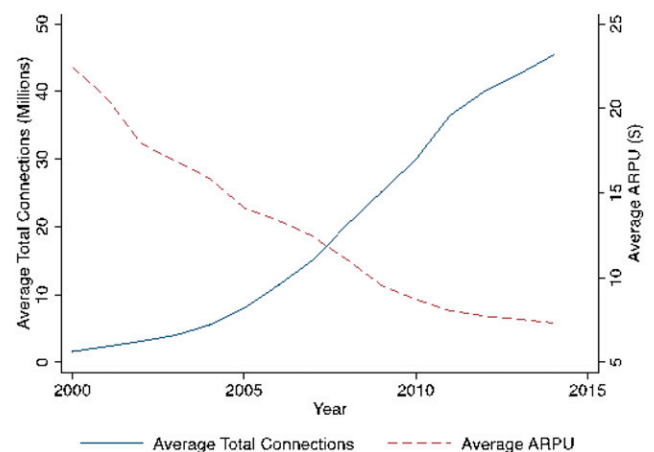
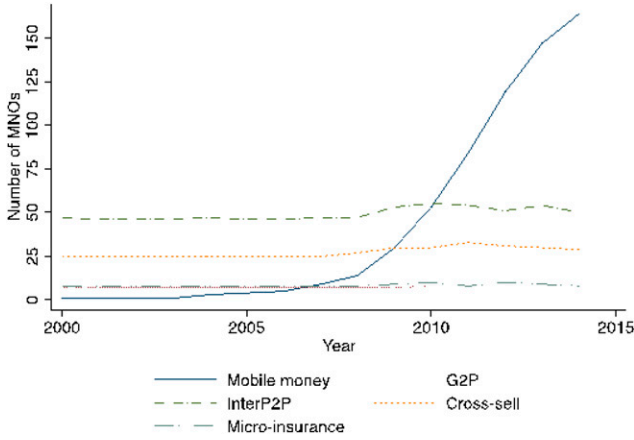


Figure 2. (Color online) Mobile Money and Bundled Services



whereas Tanzania has the highest percentage of mobile money users.

Similar to McKie et al. (2018), we use aggregate data for our estimations. We aggregated the first data set to quarterly level and merged it with the second data set by using country and MNO fields. In particular, we aggregated the data (i.e., technology, mobile money, bundled services, etc.) in each country and quarter by MNO. As a result, the compiled data for analysis is panel data with MNO as the unit and quarter as the time and has 12,656 observations. Since GDP per capita is a country-level variable, we have 3,369 unique values of this variable in our final data, with the United Arab Emirates having the maximum value. We measure mobile money using a binary variable, wherein 1 denotes an MNO offered mobile money and 0 otherwise. Similarly, for the bundled services (i.e., G2P, InterP2P, cross-sell, and microinsurance), each of the services is coded using a binary variable with 1 denoting the service was offered and 0 otherwise. Mobile technology is measured as a categorical variable with four categories (1G, 2G, 3G, and 4G). Table 1 presents the definitions and descriptive statistics of all variables.

3.2. Econometric Model

We apply the Berry (1994) method for demand estimation in our paper. The method has been widely used in economics, marketing, and information systems fields (e.g., Berry 1994, Berry et al. 1995, Kadiyali et al. 2001, Narayanan et al. 2005, Ghose and Han 2014). This model allows us to estimate the impact of price of technology and mobile money on demand using aggregate data. The utility of an individual i purchasing service j is

$$U_{ij} = \alpha p_j + \beta \mathbf{x}'_j + \xi_j + \epsilon_{ij}, \quad (1)$$

where p_j is the average price (ARPU) of purchasing service j , $\mathbf{x}'_j$ is a vector of characteristics observed by

both the researchers and consumers, ξ_j is a vector of characteristics (i.e., service characteristics, environmental characteristics, demographic characteristics, etc.) unobserved by the researchers but observed by consumers, and ϵ_{ij} is an error term representing consumer i 's idiosyncratic preferences for service j . Then, the aggregate utility for service j is

$$\delta_j = \alpha p_j + \beta \mathbf{x}'_j + \xi_j. \quad (2)$$

Using Berry's (1994) inversion method, we obtain:

$$\ln(s_j) - \ln(s_0) = \delta_j = \alpha p_j + \beta \mathbf{x}'_j + \xi_j, \quad (3)$$

where s_j is the market share of service j and s_0 is the market share of the outside option. We compute s_j as the ratio of "Total Connections" for a service in a country in a quarter to the total population in that country in that quarter. For example, Indosat in Indonesia in the first quarter of 2006 had 12,955,680 total connections, and Indonesia's population in the first quarter of 2006 was 226,902,591, thus, Indosat had a market share of 0.06 (6%). s_0 is computed as the ratio of the difference between total population in a country in a quarter and the sum of Total Connections for all services in that country in that quarter to the total population in that country in that quarter. For example, the market share of the outside option (s_0) in Indonesia in the first quarter of 2006 was 0.78 (78%). It should be noted that the choice of the outside option will not affect our results, qualitatively, because the magnitude of the outside option (and therefore, its market share) would affect the dependent variable, in Equation (3), for all observations in a given market by the same amount quantitatively.

The nonnested model described above violates the property of independence of irrelevant alternatives in our setting. For example, customers who use 3G technology may be more likely to choose another MNO that provides 3G technology when their first

Figure 3. (Color online) Total Connections, in Percentage, by Mobile Technologies

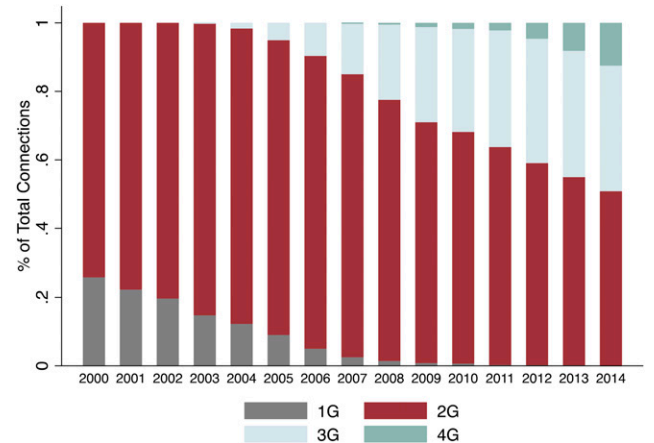
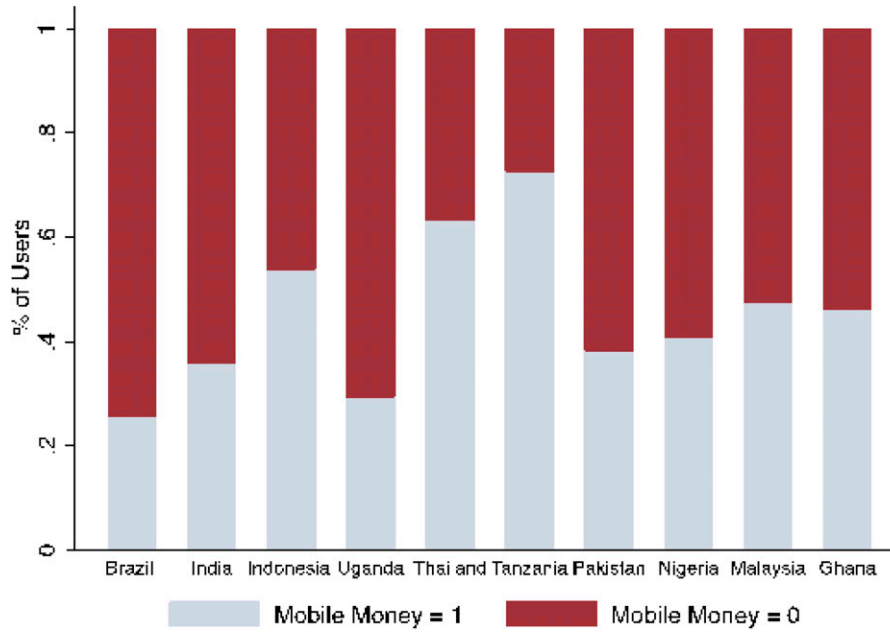


Figure 4. (Color online) Mobile Money in Top 10 Countries

MNO choice is unavailable, or customers who use mobile money service may be more likely to choose another MNO that provides mobile money service when their first choice is unavailable. Thus, in order to estimate the consumer demands for different technologies and different services, we build a number of models in which we nest our data either by technology (e.g., 3G) or by service (e.g., mobile money).

For the technology nested model, we create five nests ($g = 0, 1, 2, 3, 4$) based on technology offering where $g = 1$ contains 1G services, $g = 2$ contains 2G

services, $g = 3$ contains 3G services, $g = 4$ contains 4G services offerings, and $g = 0$ contains the outside option. The nested logit model is derived using the Berry (1994) inversion method:

$$\ln(s_j) - \ln(s_0) = \alpha p_j + \beta x'_j + \sigma \log(s_{j|g}) + \xi_j, \quad (4)$$

where σ is the substitutability factor ($0 < \sigma < 1$) and $s_{j|g}$ is the market share of service j in nest g .

Since our main model is nested by country, we do not add MNO fixed effects or country fixed effects to

Table 1. Definitions and Descriptive Statistics of All Variables

Variable	Definition	Mean	Standard deviation	Minimum	Maximum
ARPU (\$)	Average revenue per user	11.63	10.10	0.08	116.04
Total connections	Total number of unique SIM cards registered on a mobile network at the end of a quarter	4.77×10^6	1.42×10^7	26	1.99×10^8
GDP per capita (\$)	Ratio of GDP at purchaser's price to population	1750.90	3069.59	74.27	34,745.70
Population	Total number of all residents	1.39×10^8	3×10^8	588,309	1.27×10^9
Mobile money	Dummy variable indicating whether mobile money was offered for an MNO during a quarter	0.15	0.36	0	1
G2P	Dummy variable indicating whether G2P service was offered for an MNO during a quarter	0.02	0.15	0	1
InterP2P	Dummy variable indicating whether international P2P payment service was offered for an MNO during a quarter	0.19	0.39	0	1
Cross-sell	Dummy variable indicating whether selling related or complementary products to an existing user was offered for an MNO during a quarter	0.08	0.27	0	1
Microinsurance	Dummy variable indicating whether any microinsurance product was offered for an MNO during a quarter	0.03	0.18	0	1

Note. ARPU, average revenue per user; GDP, gross domestic product; MNO, mobile network operator.

avoid any potential multicollinearity issues. However, as robustness checks (see Section 4.4), we include country and MNO fixed effects and obtain consistent results.

3.3. Estimation Procedure

Price (ARPU) and nested market share are potentially endogenous because they may be correlated with unobserved covariates. Further, the decision of an MNO to offer services such as mobile money and other bundled services may also be potentially endogenous. The Durbin χ^2 statistic of 99.749 ($p < 0.001$) and the Wu-Hausman F statistic of 50.219 ($p < 0.001$) confirms the presence of the endogeneity, statistically. Therefore, we use instrumental variables (IVs) for these endogenous variables, which is a standard treatment (e.g., Guajardo et al. 2016, McKie et al. 2018). Specifically, we use the sum of the characteristics (including the sum of dummy variables for mobile money and all the bundled services) of other observations within a nest in a market as instrumental variables. The sum of the characteristics of other services are appropriate instrumental variables because they are excluded from the utility equation (U_{ij} or δ_j does not depend on the characteristics of other observations) but correlated with prices via the markups in the first-order conditions (Berry 1994, Berry et al. 1995). A similar set of instruments have been used in prior information systems and operations management studies that use aggregate choice models (e.g., Ghose and Han 2014, Guajardo et al. 2016, McKie et al. 2018). We use two-stage least squares (2SLS) method to estimate our model (Angrist and Krueger 1994).

In the first stage of 2SLS, the null hypothesis that the excluded instruments have no explanatory power is rejected (Staiger and Stock 1997). In particular, the F -statistics for all endogenous variables are much greater than the recommended threshold value of 10, demonstrating the strength of our instrumental variables. Second, we run the test of under-identification of instruments. The null hypothesis that our instruments are under-identified is also rejected (the Anderson Canonical Correlation LM Statistic was 340.645 ($p < 0.001$)). These test results demonstrate the validity of our model and the use of instrumental variables to address the endogeneity (Guajardo et al. 2016, McKie et al. 2018).

4. Analysis and Results

4.1. Demand for Mobile Technology

The estimation results for the technology nested model with instruments are presented in Table 2. As can be gleaned, the coefficient of price is negative and significant ($\beta = -0.174$, $p < 0.001$). The coefficient of nested market share is positive and significant

Table 2. Demand Estimation by Mobile Technology

Variable	Coefficient	Standard error
Price	-0.174***	(0.017)
$\log(s_{j g})$	0.764***	(0.020)
GDP per capita	3.994×10^{-4} ***	(2.520×10^{-5})
Mobile money	0.781***	(0.127)
G2P	-0.743***	(0.179)
InterP2P	0.030	(0.082)
Cross-sell	-0.520*	(0.249)
Microinsurance	-0.708***	(0.143)
Intercept	-0.760***	(0.190)
Observations	12,537	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter fixed effects.
*** $p < 0.001$; * $p < 0.05$.

($\beta = 0.764$, $p < 0.001$) and in the acceptable range of 0 to 1 in order to support nesting (Berry 1994). Further, the coefficient of mobile money is positive and significant ($\beta = 0.781$, $p < 0.001$), whereas the coefficients of G2P ($\beta = -0.743$, $p < 0.001$), cross-sell ($\beta = -0.520$, $p < 0.05$), and microinsurance ($\beta = -0.708$, $p < 0.001$) are negative and significant.

Next, we use the estimates given in Table 2 and Equation (5) to compute own-price and cross-price elasticities for the technology-nested model.

$$\frac{\partial s_i}{\partial p_j} \frac{p_j}{s_i} = \begin{cases} \alpha p_i \left[\frac{1}{1-\sigma} - \left(\frac{\sigma}{1-\sigma} \right) s_{j|g} \right] & \text{if } i = j, \\ -\alpha p_i s_i & \text{if } i \neq j, i \in g, j \notin g \\ -\alpha p_j \left[\left(\frac{\sigma}{1-\sigma} \right) s_{j|g} + s_j \right] & \text{if } i \neq j, i, j \in g, \\ -\alpha p_j s_j & \end{cases} \quad (5)$$

where i, j index observations.

Table 3 shows the own-price and cross-price elasticities for all technologies. The diagonal numbers are average own-price elasticities for each of the four technologies, and the off-diagonal numbers are average cross-price elasticities for each technology pair. For the own-price elasticities shown in Table 3, we test whether the own-price elasticities of technologies g ($g = 2, 3, 4$) are significantly different than the own price elasticity of technology 1G at the 95% significance level by using McKie et al.'s (2018) approach.

Table 3. Own and Cross-Price Elasticities by Mobile Technology

Technology	1G	2G	3G	4G
1G	-5.68	1.34	2.21	—
2G	2.42	-6.19	1.87	3.57
3G	1.66	1.65	-4.56 ^a	3.27
4G	—	2.30	2.44	-5.81

^aDenotes own-price elasticity of technology g ($g = 2, 3, 4$) is different from own-price elasticity of technology 1G at 95% significance level.

For this, we use the upper and lower confidence interval estimates of α and σ to estimate our significance level by checking to see if the elasticity estimates (calculated using either the upper or lower values from the 95% confidence intervals) have any overlaps. The own-price elasticity shows the resulting percent increase in market share when the price of a technology decreases by 1%. For example, a 1% decrease in the price of an MNO's 3G technology increases its market share by 4.56%, on average. Among all technologies, we find that the 3G technology is the least sensitive to changes in its own-price because its own-price elasticity is significantly lower than other technologies' own-price elasticities. This is likely due to the fact that 3G technology is the current, dominant technology that has robust demand as compared with other technologies that are either declining (i.e., 1G and 2G) or new to the market (i.e., 4G). There are no significant differences in the own-price elasticities among 1G, 2G, and 4G technologies.

The cross-price elasticities in Table 3 show how price changes in the technologies listed in the columns affect market share of the technologies listed in the rows.⁷ For example, a 1% decrease in the price of 2G (second column) decreases the market shares for 1G by 1.34%, 3G by 1.65%, and 4G by 2.30%. There are two notable patterns. First, the cross-price elasticities for 4G-2G and 4G-3G pairs are 3.57 and 3.27, respectively, higher than those for other pairs, indicating that the losses in the market shares of 2G and 3G technologies are higher when the price of 4G technology decreases as compared with similar decreases in the price of other technologies. Second, the cross-price elasticities are higher for the effects of a technology on newer technology than on older technologies. In particular, when the price of 2G decreases by 1%, the demand for 1G (older technology) decreases by 1.34%, whereas those for 3G and 4G (newer technologies) decrease by 1.65% and 2.30%, respectively, and when the price of 3G decreases by 1%, the demands for 1G and 2G (older technologies) decrease by 2.21% and 1.87%, respectively, and that for 4G (newer technology) decreases by 2.44%.

4.2. Effect of Mobile Money

To estimate the effect of mobile money, we assume that customers are more likely to choose an MNO that provides mobile money when their first choice is unavailable. Therefore, we nest our model on the basis of provision of mobile money. More specifically, we re-estimate Equation (4) by nesting our observations on the basis of mobile money. The estimation results are presented in Table 4. As can be gleaned from Table 4, the coefficient of price is negative and significant ($\beta = -0.069$, $p < 0.001$). Also, the coefficient of nested market share is positive and significant ($\beta = 0.917$,

Table 4. Demand Estimation for Mobile Money

Variable	Coefficient	Standard error
Price	-0.069***	(0.002)
$\log(s_{j g})$	0.917***	(0.009)
GDP per capita	2.766×10^{-4} ***	(5.800×10^{-6})
2G Technology ^a	1.182***	(0.066)
3G Technology ^a	2.383***	(0.066)
4G Technology ^a	3.292***	(0.162)
G2P	-0.249*	(0.108)
InterP2P	0.006	(0.044)
Cross-sell	0.031	(0.062)
Microinsurance	-0.560***	(0.090)
Intercept	-1.991***	(0.082)
Observations	12,537	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter fixed effects.

^aHoldout Group: 1G Technology.

*** $p < 0.001$; * $p < 0.05$.

$p < 0.001$) and is between 0 and 1, as required to support nesting (Berry 1994). Further, the provision of newer technologies (3G and 4G as compared with 1G and 2G) is associated with higher market shares.

Again, using the estimates in Table 4 and Equation (5), we estimate own-price and cross-price elasticities for the provision of mobile money. As can be seen in Table 5, the own-price elasticity with the provision of mobile money is significantly lower, in magnitude, than without the provision of mobile money. In particular, a 1% decrease in the price of the service with mobile money is associated with 4.14% increase in the market share, whereas a 1% decrease in the price of the service without mobile money is associated with 7.81% increase in the market share. For the cross-price elasticities, the change in price of the service with mobile money has greater impact on the market share of the service without mobile money than the change in price of the service without mobile money has on the market share of the service with mobile money. That is, a 1% decrease in the price of the service without mobile money decreases the market share of the service with mobile money by only 1.33%, whereas a 1% decrease in the price of the service with mobile money decreases the market share of service without mobile money by 2.22%.

In addition to the mobile money, we also estimate the own-price and cross-price elasticities for other bundled services (i.e., G2P, InterP2P, cross-sell, and microinsurance), one at a time and presented the

Table 5. Own and Cross-Price Elasticities by Mobile Money

Mobile Money	0	1
0	-7.81	2.22
1	1.33	-4.14 ^a

^aDenotes own-price elasticity of the service with mobile money is different from own-price elasticity of the service without mobile money at 95% significance level.

results in Tables A.2–A.5 in the appendix. Similar to mobile money, the own-price elasticity with the provision of a bundled service is significantly lower, in magnitude, than without the provision of the bundled service. Moreover, the change in price with the bundled service has greater impact on the market share of service without the bundled service than the change in price without the bundled service has on the market share of service with the bundled service. Hence, the demand patterns are consistent among all financial services (i.e., Mobile Money, G2P, InterP2P, cross-sell, and microinsurance), demonstrating the robustness of our models and estimations.

4.3. Effect Between Technology and Mobile Money

In order to examine whether the own-price elasticities of mobile money vary across different generations of mobile technology, we estimate the model that nested our observations on both technology and mobile money. The results are presented in Table 6. As can be gleaned from Table 6, there are two consistent patterns among all generations of mobile technology (1G, 2G, 3G, and 4G). First, the magnitude of own-price elasticity without mobile money is significantly higher than the magnitude of own-price elasticity with mobile money. For example, for 3G technology, the own-price elasticity without mobile money is -1.60 which is higher than the own-price elasticity with mobile money of -0.75 . Second, the magnitude of own-price elasticity without mobile money for 3G technology is the lowest as compared with those for other technologies (i.e., -1.60 for 3G vs. -1.96 for 1G, -2.29 for 2G and -1.86 for 4G), likely because 3G is the current, dominant technology. However, those with mobile money for 3G are at the similar level with other technologies. In other words, if an MNO provides mobile money, customers are less price-sensitive and less likely to switch to a competitor. The patterns for other bundled services are similar and consistent as well (see Appendix Tables A.6–A.9).⁸

Table 6. Own-Price Elasticities by Technology and Mobile Money

Technology	Mobile money	Own-price elasticity
1G	0: No	-1.96^a
	1: Yes	-0.67
2G	0: No	-2.29^a
	1: Yes	-0.68
3G	0: No	-1.60^a
	1: Yes	-0.75
4G	0: No	-1.86^a
	1: Yes	-0.98

^aDenotes own-price elasticity when mobile money = 0 is different from own-price elasticity when mobile money = 1 for a given technology at 95% significance level.

Table 7 presents the elasticities of mobile money and the bundled services G2P, cross-sell, and microinsurance (note that only G2P, cross-sell, and microinsurance were significant in Table 2) by each generation of technology.⁹ The estimates show how the offering of a service affect the MNO's market share for a given generation of technology. The overall effect of mobile money is 0.40%; that is, providing mobile money can increase an MNO's market share by 0.40%, on average. However, there are considerable differences across different technologies. In particular, the effect of mobile money is the lowest for 1G, followed by 2G, and the highest for 3G and 4G technologies. Contrary to the effect of mobile money, the effects of the bundled services G2P, cross-sell, and microinsurance are negative; that is, providing these bundled services is associated with lower market shares. These results corroborate asymmetric benefits findings of Dong et al. (2018).

4.4. Robustness Tests

To check the robustness of our main results with respect to the coefficients of price (ARPU) and nested market share, we performed six tests. First, we included MNO fixed effects in our model. The results are presented in Table A.10 in the appendix. The coefficients of price and nested market share are consistent with those in the main results. Second, we included country fixed effects. The results are presented in Table A.11 in the appendix. Again, the coefficients of price and nested market share are consistent. Note that these estimations may suffer from multicollinearity because we nest our observations on the basis of country as well. These estimations and results are only for the purpose of checking the robustness of our main results with respect to the coefficients of ARPU and nested market share. Third, we used GDP growth, instead of GDP per capita, as a control variable. As shown in Table A.12 in the appendix, the coefficients of price and nested market share remain consistent.

Fourth, customers may use mobile money and the bundled services because they do not have access to other services such as credit cards. In other words, customers may more likely choose mobile money when credit card diffusion is less in a certain country. To account for this, we controlled for credit card diffusion in each country in our model. Although we were not

Table 7. Elasticities of Mobile Money and Bundling Services: Nested on Technology

Technology	Mobile money	G2P	Cross-sell	Microinsurance
1G	0.02	-0.02	-0.01	-0.02
2G	0.32	-0.06	-0.12	-0.06
3G	0.72	-0.05	-0.11	-0.05
4G	0.64	-0.01	-0.03	–
Overall	0.40	-0.05	-0.11	-0.05

able to collect the credit card diffusion data for all years in our data, we collected the data for 2011 and 2014. Therefore, we used the average credit card diffusion between 2011 and 2014 as the credit card diffusion for all quarters in our data and added it as a control variable in our estimation. The results are presented in Table A.13 in the appendix and again, the coefficients of price and nested market share remain consistent.

Fifth, Brazil and India have experienced higher growth rates among emerging economies and may have different demand patterns from other countries. Therefore, we added two dummies for these two countries in our model. As shown in Table A.14 in the appendix, our main results remain consistent.

Sixth, we estimated an alternate model by aggregating our data to country level and estimating the model with country fixed effects. Because we aggregated all our observations to the country level, we were not able to estimate the nested logit model because the nested market share is 1, by definition, for all revised observations. Thus, we estimated a nonnested logit model and included technology as an additional control instead of treating it as a nesting variable. The results are presented in Table A.15 in the appendix and again, the main results remain consistent.

5. Concluding Remarks

Mobile money is an innovative financial technology bundled with both basic and value-added financial services and supported by multiple generations of mobile technologies. It has been successful in providing financial inclusion for the under-served populations in the developing world. As mobile technologies advance and competition increases, MNOs, however, are challenged with adapting to newer technologies and adjusting service offerings. Using a data set on mobile money services and mobile network operations in the emerging markets compiled from multiple sources, we examine the demand patterns of mobile technologies and mobile money. Using a structural model, we estimate the own- and cross-price elasticities for both mobile technology and mobile money, and the demand effects between mobile money and mobile technologies. We also extend the estimations to other financial services such as G2P, cross-sell, microinsurance, and international P2P.

We find that 3G technology has a significantly smaller own-price elasticity than other generations of mobile technology, suggesting that 3G technology is least sensitive to changes in its own price. This is likely due to the fact that 3G technology was the current technology in demand, thereby having more robust demand. Interestingly, when the price increases for 1G, 2G, and 3G technologies, customers switch to other technologies by substituting forward. That is, more customers are to purchase newer technology when the price for

the technology they use increases. We also find that mobile money mitigates the competition among MNOs by making customers less sensitive to the mobile technologies. When mobile money is added to the technology-service bundle, the elasticity is greatly reduced, suggesting that consumers are less likely to abandon the service or to defect from the MNO. By adopting mobile money, MNOs can effectively differentiate the market with an innovative mobile service, which reduces the intensity of price competition for these MNOs but increases the strengths against those without mobile money. Furthermore, we find that mobile money increases the market share of an MNO for all generations of mobile technology, whereas the newer technologies (3G and 4G) increase the most.

We also find that bundling mobile money with other financial services such as G2P, cross-sell, and microinsurance may negatively affect the market share. This suggests that the tradeoff between the benefits of service bundling, such as more consumer choices of services and more deeply differentiated market, and the costs may lead to a balanced bundling strategy for MNOs, which varies with the technology platform that supports the services. It is important to note, however, despite the negative effects of G2P, cross-sell, and microinsurance bundles, the positive effect of mobile money continue to dominate so that the combined effects on market share remain positive.

Our findings have important managerial implications. First, although mobile money has been purported as a technological innovation to promote financial inclusion for the under-served populations, our findings indicate that it also benefits MNOs. Hence, it is important for managers to understand the role of mobile money in both providing social good and making a profit. Second, the effect of mobile money remains positive as mobile technology evolves and mobile capacity increases. From 1G to 4G, mobile money consistently serves as a competitive advantage for MNOs. However, this does not mean that the effect stays unchanged over the generations of mobile technology. Instead, when the 3G technology allows web browsing as a major upgrade of mobile functionality, MNOs with mobile money have a substantially larger set of options to differentiate from those without mobile money; and as a result, mobile money implemented with 3G and 4G leads to larger market shares than that with 1G and 2G. Third, although service bundling with mobile money has benefits for users, it comes with costs. In anticipation of competitive threats, MNOs may decide to offer additional value-added services such as G2P and microinsurance, on top of the basic services such as airtime and domestic P2P. However, we find that bundling mobile money with G2P and microinsurance may trim down the benefit of mobile money.

This may indicate that services without a clear value proposition for customers, for example, services that are too costly or complex to use, may cause diminishing returns for MNOs when they continue to add options. The results highlight that there may exist an optimal level of service bundling for MNOs in offering mobile money services.

Our study has a few limitations and we believe future research, addressing our limitations and exploring the following avenues, can contribute further to both theory and practice on mobile money. First, our data are for emerging markets. Hence, the findings should be generalized with caution. Nevertheless, although limited for emerging markets, our final data (used for all estimations) includes 68 countries that had a combined GDP of over 4 trillion in 2014. Second, due to data limitations, we use a binary measure for mobile money and other bundled services. It would be ideal if we could measure the usage of mobile money and other financial services. Future

research can collect the more granular usage data and estimate whether the patterns of consumer demand hold in general. Third, our analyses are at an aggregate level. Using consumer-level data, future research could explore our research questions at a more granular level that would capture the demographic differences. Future research could also examine the effects of other bundling services, such as utility services and cash disbursement services, on consumer demand and price elasticities.

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All authors contributed equally to the paper.

Appendix

Table A.1. List of All Countries and MNOs in Source Data

Country	MNOs
Afghanistan	Afghan Telecom, Afghan Wireless (TSI), Etisalat, MTN, Roshan (TDCA), Wasel Telecom
Armenia	Beeline (VimpelCom), Orange, VivaCell (MTS)
Bahrain	Batelco, Viva (Saudi Telecom), Zain
Bangladesh	Airtel (Bharti Airtel), Banglalion Communications, Citycell (Pacific Bangladesh), Grameenphone (Telenor), Qubee (Augere), Robi (Axiata), Teletalk, Banglalink (Global Telecom)
Benin	Bell Benin Communications, Glo Mobile (Globacom), Libercom (Benin Telecoms), MTN, Moov (Maroc Telecom)
Botswana	Be Mobile (BTC), Mascom (MTN), Orange
Brazil	Algar Telecom, Amazonia, Brasil Telecom, Claro (América Móvil), Claro- ATL/ Algar, Claro Americel, Claro BCP, Claro BSE, Claro-Telet, Claro Tess, Nextel (NII), Oi, Sercomtel, TIM (Telecom Italia), TIM Celular, TIM Maxitel, TIM TCS, TIM TNC, Telemig, Vivo (Telefónica), Vivo-CRT, Vivo TCO, Vivo-Tele Leste, Vivo-Tele Sudeste, Vivo TeleSP, Vodacom, aeioi (Unicel)
Burundi	Econet Wireless, ONAMOB (ONATEL), Smart, TEMPO (VTEL), leo (Econet)
Cameroon	CamTel, MTN, Nexttel (Viettel), Orange
Chad	Airtel (Bharti Airtel), Sotel, Tchad Mobile, Tigo (Millicom)
Chile	BellSouth, Claro (América Móvil), Entel, Interexport, Movistar (Telefónica), Nextel (Novator), VTR Movil (Liberty Global)
Colombia	Avantel, Claro (América Móvil), ETB, Movistar (Telefónica), Tigo (Millicom), UNE (EPM)
Congo	Airtel (Bharti Airtel), Azur (Bintel), Cyrus, MTN, Warid Telecom (Abu Dhabi)
Congo Democratic Republic	Africell (Lintel), Airtel (Bharti Airtel), Orange, Starcel, Supercell, Tatem Telecom, Tigo (Millicom), Vodacom
Dominican Republic	Claro (América Móvil), Orange (Altice), Tricom, Viva (Trilogy)
Egypt	Etisalat, Mobinil (Orange), Vodafone
El Salvador	Claro (América Móvil), Digicel, Movistar (Telefónica), Red (Intelfon), Tigo (Millicom)
Ethiopia	Ethio telecom
Fiji	Digicel, Vodafone (ATH)
Gabon	Airtel (Bharti Airtel), Azur (Bintel), Libertis (Gabon Telecom), Moov (Maroc Telecom)
Ghana	Airtel (Bharti Airtel), Blu, Econet Wireless, Expresso (Sudatel), Glo Mobile (Globacom), MTN, Surfline, Tigo (Millicom), Vodafone
Guatemala	BellSouth, Claro (América Móvil), Movistar (Telefónica), Red (Intelfon), Tigo (Millicom)
Guinea	Cellcom, Intercel+ (Sudatel), Orange (Sonatel), SOTELGUI, Spacetel, areeba (MTN)
Guinea-Bissau	Guinetel (Guine Telecom), MTN, Orange (Sonatel)
Haiti	Digicel, HaiTel, Natcom (Viettel), Voila (Comcel)

Table A.1. (Continued)

Country	MNOs
Honduras	Claro (América Móvil), Digicel, Hondutel, Tigo (Millicom)
India	Aircel (Maxis), Airtel (Bharti Airtel), BSNL, Cheers Mobile (Etisalat DB), Hexacom, Idea Cellular, Loop Mobile, MTNL, MTS (Sistema Shyam TeleServices), Quadrant Televentures, Reliance Communications, S Tel, Skycell, Spice Telecom, Tata DOCOMO (Tata Teleservices), Uninor (Telenor), Videocon Telecom, Vodafone
Indonesia	3 (Hutchison), Axis (Saudi Telecom), BOLT! (Internux), Ceria (Sampoerna Telekomunikasi), Indosat (Ooredoo), Komselindo, Metrosel, Smart Telecom, Smartfren, Telesera, Telkomsel (Telekomunikasi Indonesia), XL (Axiata)
Iran	MCI (TCI), MTCE, MTN Irancell, Rightel (Tamin Telecom), TKC, Taliya
Jordan	Orange (Jordan Telecom), Umniah (Batelco), Xpress, Zain, wi-tribe
Kenya	Airtel (Bharti Airtel), Orange (Telkom Kenya), Safaricom, yuMobile (Essar Telecom)
Lesotho	Econet Wireless, Vodacom
Liberia	Cellcom, LIBTELCO, LiberCell (Atlantic Wireless), Lonestar (MTN), Novafone
Madagascar	Airtel (Bharti Airtel), Intercel, Life (Madamobil), Orange, Telma
Malawi	Airtel (Bharti Airtel), TNM
Malaysia	Celcom (Axiata), DiGi, Electcoms, Maxis, P1 (Telekom Malaysia), Telekom Cellular, Telekom Malaysia, U Mobile, Yes (YTL Communications)
Mali	Malitel (Sotelma), Orange (Sonatel)
Mauritania	Chinguitel (Sudatel), Mattel (Tunisie Telecom), Mauritel
Mauritius	Emtel (Millicom), MTML, Orange (Mauritius Telecom)
Mexico	Axtel, Iusacell (AT&T), Movistar (Telefónica), Nextel (NII), Telcel (América Móvil), Unefon
Mongolia	G-Mobile, MobiCom, Skytel, Unitel
Morocco	Maroc Telecom (Etisalat), Meditel, inwi (Wana)
Mozambique	Movitel (Viettel), Vodacom, mcel (Mozambique Cellular)
Namibia	MTC, TN Mobile (Telecom Namibia), leo (Powercom)
Nepal	Hello Nepal (Nepal Satellite Telecom), Ncell (TeliaSonera), Nepal Telecom, Smart Cell (Smart Telecom), United Telecom
Niger	Airtel (Bharti Airtel), Moov (Maroc Telecom), Orange, SahelCom (Sonitel)
Nigeria	Airtel (Bharti Airtel), Etisalat (EMTS), Glo Mobile (Globacom), MTN Mtel (NITEL), Multi-Links (Capcom), Smile, Starcomms, Visafone, ZOOMmobile (RelTel Wireless)
Pakistan	Instaphone (Pakcom), Mobilink (Global Telecom), PTCL, Qubee (Augere), Special Communications Organization, Telenor, Warid Telecom (Abu Dhabi), Wateen Telecom (Abu Dhabi), Zong (China Mobile), wi-tribe (Ooredoo)
Panama	Cable & Wireless, Claro (América Móvil), Digicel, Movistar (Telefónica)
Papua New Guinea	Citifon (Telikom PNG), Digicel, bmobile
Paraguay	Claro (América Móvil), Personal (Telecom Argentina), Tigo (Millicom), Vox (Hola)
Peru	BellSouth, Bitel (Viettel), Claro (América Móvil), Entel, Movistar (Telefónica)
Philippines	Express Telecom, Globe Telecom, Now Telecom, Smart (PLDT), Sun Cellular (Digitel), Talk 'N Text, wi-tribe (Liberty Telecom)
Qatar	Ooredoo, Vodafone
Romania	DigiMobil (RCS & RDS), Orange, Telekom Romania (OTE), Vodafone, Zapp
Rwanda	Airtel (Bharti Airtel), MTN, RwandaTel (Lap Green Network), Tigo (Millicom)
Samoa	BlueSky Communications, Digicel
Senegal	Expresso (Sudatel), Orange (Sonatel), Tigo (Millicom)
Sierra Leone	Africell (Lintel), Airtel (Bharti Airtel), Comium, SierraTel, Tigo (Millicom)
Singapore	Grid Communications, M1, Singtel, StarHub
Somalia	Barakaat, Golis Telecom, Hormuud Telecom, NationLink Telecom, Somafone Telcom, Telesom Somaliland
South Africa	Cell C (Oger Telecom), MTN, Telkom Mobile, Vodacom
Sri Lanka	Airtel (Bharti Airtel), Dialog (Axiata), Etisalat, Hutch (Hutchison), Mobitel (Sri Lanka Telecom)
Sudan	MTN, Sudani (Sudatel), Zain
Swaziland	MTN
Tanzania	Airtel (Bharti Airtel), BOL Mobile (Benson Informatics), Sasatel (Dovetel), Smart, Smile, TTCL, Tigo (Millicom), TriTel, Vodacom, Zantel (Etisalat)
Thailand	AIS, CAT, DTAC (Telenor), Digital Phone Company, Hutch, TOT, True Move (True Corporation), WE PCT (Asia Wireless Communication), my (CAT Telecom)
Timor-Leste	Telkomcel (Telin), Telmor (Viettel), Timor Telecom
Togo	Moov (Maroc Telecom), Togocel (Togo Telecom)
Tonga	Digicel, U-Call (Tonga Telecom)
Tunisia	Ooredoo (NMT), Orange, Tunisie Telecom
Turkey	Avea (Türk Telekom), Aycell, Turkcell, Vodafone
Uganda	Africell (Lintel), Airtel (Bharti Airtel), MTN, Orange, Smart, Smile, UT Mobile (Uganda Telecom), Vodafone (Afrimax), Warid Telecom (Abu Dhabi), i-Tel
United Arab Emirates	Etisalat

Table A.1. (Continued)

Country	MNOs
Vanuatu	Digicel, Smile (Telecom Vanuatu)
Vietnam	E-Mobile (EVNTelecom), Gmobile (GTEL Mobile), Mobifone, S-Fone (S-Telecom), Shoreline, Vietnamobile (Hanoi Telecom / Hutchison), Viettel Telecom, Vinaphone (VNPT)
Zambia	Airtel (Bharti Airtel), MTN, Zamtel
Zimbabwe	Econet Wireless, NetOne, Telecel (Global Telecom)
Cote d'Ivoire	Caf Mobile (Aircel), Comstar, GreenN (Oricel), MTN, Moov (Maroc Telecom), Orange, YooMee, koZ (Comium)

Table A.2. Own and Cross-Price Elasticities by G2P

G2P	0	1
0	−3.26	0.80
1	0.16	−1.35 ^a

^aDenotes own-price elasticity of service with G2P is different from own-price elasticity of service without G2P at 95% significance level.

Table A.3. Own and Cross-Price Elasticities by InterP2P

InterP2P	0	1
0	−2.84	1.51
1	0.97	−2.13 ^a

^aDenotes own-price elasticity of service with InterP2P is different from own-price elasticity of service without InterP2P at 95% significance level.

Table A.4. Own and Cross-Price Elasticities by Cross-sell

Cross-sell	0	1
0	−3.54	1.64
1	0.95	−1.32 ^a

^aDenotes own-price elasticity of service with cross-sell is different from own-price elasticity of service without cross-sell at 95% significance level.

Table A.5. Own and Cross-Price Elasticities by Microinsurance

Microinsurance	0	1
0	−3.37	1.14
1	0.36	−0.73 ^a

^aDenotes own-price elasticity of service with microinsurance is different from own-price elasticity of service without microinsurance at 95% significance level.

Table A.6. Own-Price Elasticities by Technology and G2P

Technology	G2P	Own-price elasticity
1G	0: No	−1.20 ^a
	1: Yes	−0.27
2G	0: No	−1.21 ^a
	1: Yes	−0.66
3G	0: No	−0.92 ^a
	1: Yes	−0.25
4G	0: No	−1.21 ^a
	1: Yes	−0.07

^aDenotes own-price elasticity when mobile money = 0 is different from own-price elasticity when mobile money = 1 for a given technology at 95% significance level.

Table A.7. Own-Price Elasticities by Technology and InterP2P

Technology	InterP2P	Own-price elasticity
1G	0: No	−1.36 ^a
	1: Yes	−0.76
2G	0: No	−1.27 ^a
	1: Yes	−0.74
3G	0: No	−0.93 ^a
	1: Yes	−0.63
4G	0: No	−1.20 ^a
	1: Yes	−0.81

^aDenotes own-price elasticity when mobile money = 0 is different from own-price elasticity when mobile money = 1 for a given technology at 95% significance level.

Table A.8. Own-Price Elasticities by Technology and Cross-sell

Technology	Cross-sell	Own-price elasticity
1G	0: No	−1.34 ^a
	1: Yes	−0.83
2G	0: No	−1.38 ^a
	1: Yes	−0.56
3G	0: No	−1.05 ^a
	1: Yes	−0.36
4G	0: No	−1.38 ^a
	1: Yes	−0.42

^aDenotes own-price elasticity when mobile money = 0 is different from own-price elasticity when mobile money = 1 for a given technology at 95% significance level.

Table A.9. Own-Price Elasticities by Technology and Microinsurance

Technology	Microinsurance	Own-Price Elasticity
1G	0: No	−1.22 ^a
	1: Yes	−0.41
2G	0: No	−1.25 ^a
	1: Yes	−0.33
3G	0: No	−0.95 ^a
	1: Yes	−0.18
4G	0: No	−1.24
	1: Yes	−

^aDenotes own-price elasticity when mobile money = 0 is different from own-price elasticity when mobile money = 1 for a given technology at 95% significance level.

Table A.10. Technology Nested Model with MNO Fixed Effects

Variable	Coefficient	Standard error
Price	−0.097***	(0.003)
$\log(s_{jlg})$	0.279***	(0.023)
GDP per capita	2.372×10^{-4} ***	(1.260×10^{-5})
Mobile money	1.221***	(0.067)
G2P	0.774**	(0.242)
InterP2P	−0.287*	(0.122)
Cross-sell	−0.688*	(0.340)
Microinsurance	−0.534**	(0.171)
Intercept	−2.896***	(0.353)
Observations	12,537	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter and MNO fixed effects.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.11. Technology Nested Model with Country Fixed Effects

Variable	Coefficient	Standard error
Price	−0.328***	(0.020)
$\log(s_{jlg})$	0.499***	(0.023)
GDP per capita	5.100×10^{-5}	(4.540×10^{-5})
Mobile money	0.271*	(0.137)
G2P	−0.181	(0.199)
InterP2P	0.263*	(0.105)
Cross-sell	0.236	(0.243)
Microinsurance	0.179	(0.161)
Intercept	−0.108	(0.355)
Observations	12,537	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter and country fixed effects.

*** $p < 0.001$; ** $p < 0.05$.

Table A.12. Technology Nested Model with GDP Growth as a Control

Variable	Coefficient	Standard error
Price	−0.114***	(0.016)
$\log(s_{jlg})$	0.780***	(0.022)
GDP growth	0.237	(0.436)
Mobile money	0.781***	(0.128)
G2P	−0.729***	(0.185)
InterP2P	0.209*	(0.091)
Cross-sell	−1.599***	(0.254)
Microinsurance	−0.921***	(0.155)
Intercept	−0.612**	(0.224)
Observations	11,844	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter fixed effects.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.13. Technology Nested Model with Credit Card Diffusion as a Control

Variable	Coefficient	Standard error
Price	−0.174***	(0.016)
$\log(s_{jlg})$	0.765***	(0.020)
GDP per capita	3.991×10^{-4} ***	(2.620×10^{-5})
Mobile money	0.780***	(0.125)
G2P	−0.783***	(0.179)
InterP2P	0.031	(0.082)
Cross-sell	−0.518*	(0.249)
Microinsurance	−0.707***	(0.143)
Credit card diffusion	2.707×10^{-4}	(0.003)
Intercept	−0.759***	(0.189)
Observations	12,537	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter fixed effects.

*** $p < 0.001$; * $p < 0.05$.

Table A.14. Technology Nested Model with Country Dummies for Brazil and India

Variable	Coefficient	Standard error
Price	−0.271***	(0.022)
$\log(s_{jlg})$	0.607***	(0.023)
GDP per capita	5.313×10^{-4} ***	(3.220×10^{-5})
Mobile money	0.369*	(0.156)
G2P	−0.555**	(0.204)
InterP2P	0.157	(0.095)
Cross-sell	−1.680***	(0.287)
Microinsurance	−1.145***	(0.168)
Intercept	0.342	(0.251)
Observations	12,537	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter fixed effects and country dummies for Brazil and India.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.15. Nonnested Model with Data Aggregated to Country Level

Variable	Coefficient	Standard error
Price	−0.392***	(0.037)
GDP per capita	2.220×10^{-5}	(2.640×10^{-5})
Mobile money	0.396*	(0.195)
G2P	6.131***	(0.981)
InterP2P	5.917 ^a	(3.396)
Cross-sell	−14.013***	(5.076)
Microinsurance	−17.690 ^a	(10.222)
2G Technology ^a	6.473***	(0.131)
3G Technology ^a	2.091***	(0.122)
4G Technology ^a	−1.315***	(0.203)
Intercept	−5.586***	(0.632)
Observations	7,379	

Note. IV Estimates [DV: $\ln(s_j) - \ln(s_0)$]. Includes quarter and country fixed effects.

^aHoldout Group: 1G Technology.

*** $p < 0.001$; * $p < 0.05$.

Endnotes

- ¹ See https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2015/03/SOTIR_2014.pdf (accessed July 2019).
- ² See https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2013/06/2013_MMU_Platforms-for-Successful-Mobile-Money-Services.pdf (accessed July 2019).
- ³ See <https://www.technologyreview.com/s/603770/the-5g-economy-how-5g-will-impact-global-industries-the-economy-and-you/> (accessed July 2019).
- ⁴ See https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2013/06/2013_MMU_Platforms-for-Successful-Mobile-Money-Services.pdf (accessed July 2019).
- ⁵ See https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2013/06/2013_MMU_Platforms-for-Successful-Mobile-Money-Services.pdf (accessed July 2019).
- ⁶ Table A.1 in the appendix presents the full list of the MNOs and the countries.
- ⁷ We cannot estimate the cross-price elasticity of 1G-4G pair (and 4G-1G pair) because none of the countries in our data have both services provided in any given quarter.
- ⁸ We cannot estimate own-price elasticity for 4G technology platform with microinsurance service because none of the MNOs that provide microinsurance service has 4G technology (see Appendix Table A.9).
- ⁹ As noted above, none of the MNOs that provide microinsurance service has 4G technology in our data. Therefore, we cannot estimate the elasticity of microinsurance for 4G technology.

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